0252. Meeting Rooms

y: Easy | Category: Intervals | LeetCode: https://leetcode.com/problems/meeting-

Problem

- Given meeting time intervals, can one person attend all of them?

Intuition

- Sort by start; any meeting that starts before the previous one ends is a conflict.

Brute Force

- Compare all pairs: O(n^2).

Optimal Approach

- Sort by start.
- For i >= 1: if a[i].start < a[i-1].end return false.

Worked Example

```
[[0,30],[5,10],[15,20]].

sorted: [0,30] [5,10] [15,20]
5 < 30 -> conflict -> false
```

Java Solution

```java
public class MeetingRooms {
    public boolean canAttendMeetings(int[][] intervals) {
        int[][] a = intervals.clone();
        Arrays.sort(a, (x, y) -> Integer.compare(x[0], y[0]));
        for (int i = 1; i < a.length; i++) if (a[i][0] < a[i - 1][1]) return false;
        return true;
    }
}
```

Complexity

- Time: O(n log n). Space: O(1).

Pitfalls

- [5,10] and [10,15] do not conflict (strict <).